

CELTIC SEA BUOY 62023, Ireland

WMO: 996640

Lat: 51.400N

Lon: 7.920W

Elev: 0

StdP: 14.70

Time Zone: -1.00 (W01)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	37.8	39.5	N/A	N/A	N/A	N/A	N/A	N/A	58.1	46.1	52.4	47.1	23.9	20	0.921

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	70.2	N/A	67.6	N/A	65.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	12.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 49.0	43.7	39.5	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	47.6	46.6	47.5	50.2	53.7	58.1	61.1	62.0	60.5	56.0	51.4	48.4
	DBStd	6.35	3.12	3.05	3.24	3.18	2.99	3.60	2.90	2.35	3.09	3.20	3.40	3.59
	HDD50	395	83	98	86	31	2	0	0	0	0	2	24	69
	HDD65	4164	539	515	543	443	350	208	126	99	140	280	408	513
	CDD50	1722	9	3	8	38	117	243	345	371	315	187	66	20
	CDD65	16	0	0	0	0	0	1	6	5	4	0	0	0
	CDH74	3	0	0	0	0	0	0	0	1	2	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	20.6	25.0	22.4	19.9	19.4	18.8	17.7	17.5	17.8	18.8	21.5	23.6	25.2
(15) Precipitation	PrecAvg	40.6	4.2	3.3	2.7	2.8	2.8	2.8	3.0	3.1	3.0	4.4	4.1	4.3
	PrecMax	52.8	7.2	7.6	7.0	5.2	6.2	7.9	7.7	9.8	7.6	7.8	8.0	13.5
	PrecMin	30.1	1.3	0.6	0.6	0.5	0.1	0.5	0.7	0.3	0.5	1.1	1.1	1.1
	PrecStd	5.3	1.8	2.2	1.5	1.4	1.4	1.7	1.6	2.2	1.7	2.0	1.9	2.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	53.8	56.5	62.8	66.2	70.4	72.7	72.9	73.1	64.1	58.4	56.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	53.0	52.2	53.0	58.1	62.1	67.0	69.1	68.8	69.8	62.3	57.2	54.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	52.3	51.2	52.0	55.1	59.6	64.6	66.6	66.6	66.4	61.1	56.4	53.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	51.7	50.4	51.2	53.5	57.6	62.8	64.8	65.2	64.0	60.1	55.6	52.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.6	5.2	5.6	5.7	5.7	5.8	6.0	5.4	4.6	4.4	4.2
		MCDBR	4.8	4.5	6.1	7.8	9.2	8.7	8.1	8.0	8.8	5.5	4.7	4.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.344	0.353	0.379	0.382	0.394	0.377	0.384	0.366	0.365	0.358	0.337	0.335
	taud		2.305	2.363	2.324	2.359	2.370	2.454	2.440	2.503	2.494	2.448	2.372	2.278
	Ebn at Noon		206	242	258	272	273	279	275	275	262	238	208	186
	Edh at Noon		21	26	33	35	36	34	34	30	27	24	20	19
(44) All-Sky Solar Radiation	RadAvg		254	485	858	1408	1714	1892	1713	1451	1058	598	323	196
	RadStd		23	37	88	95	114	148	135	81	78	53	24	23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page